
ASO Azure, APIM, Functions, and Telemetry Standard

Azure resource naming, APIM policies, Function App foundation, Key Vault, telemetry, and HTTP Entra connection timing.

GitHub path: `azure/ASO_Azure_APIM_Functions_Foundation_Standard.md`

Version 1.1 - public customer-ready split package - refined for readability and GitHub publishing

Safety note: this public reference pack must not contain tenant IDs, credentials, secrets, real customer data, private endpoints, or connection strings.

Azure, APIM, Functions, Key Vault, and telemetry foundation

This file defines the Azure resource foundation and APIM/Function boundary used for SAP integration and future orchestration support. It intentionally avoids fake endpoint values. Authenticated connection references are created only when real endpoints and Entra audience values exist.

Azure naming pattern

Resource	Pattern	Example	Notes
Resource group	rg-<workload>-<env>-<region>-<###>	rg-aso-dev-weu-001	Container for ASO Azure assets.
Function App	func-<workload>-<purpose>-<env>-<region>-<###>	func-aso-sap-dev-weu-001	Hosts SAP wrapper endpoints.
API Management	apim-<workload>-<env>-<region>-<###>	apim-aso-dev-weu-001	API gateway boundary for SAP wrapper.
Key Vault	kv-<workload>-<env>-<region>-<###>	kv-aso-dev-weu-001	Store secrets; never commit secret values.
Application Insights	appi-<workload>-<env>-<region>-<###>	appi-aso-dev-weu-001	Telemetry and diagnostics.
Storage account	lowercase alphanumeric only	stasoapidevweu001	Function/runtime storage if required.

Foundation resources

- Create resource group only when the Azure subscription, tenant, and naming are approved.
- Create Application Insights early if Foundry/Functions telemetry will be linked.
- Create Key Vault before secrets are needed; never store secrets in GitHub or environment variables.
- Create Function App and APIM during the SAP/APIM phase when API contracts are defined.
- Create HTTP with Microsoft Entra ID connection references only when real base URLs and Entra resource URIs exist.

HTTP with Microsoft Entra ID connection guidance

Connection reference	Create now?	Why
ASO Foundry HTTP Entra Connection Reference	No until real endpoint/audience exists	The Base Resource URL must be the real orchestrator API endpoint; the Resource URI must be the real Entra App ID URI.
ASO APIM HTTP Entra Connection Reference	No until APIM/SAP wrapper exists	The Base Resource URL must be the real APIM URL; the Resource URI must be the protected API audience.

APIM policy requirements

- Validate JWT on inbound requests.
- Require x-correlation-id and x-message-id.

- Apply rate limits and backend routing.
- Preserve trace IDs outbound.
- Retry only safe read operations.
- Do not retry non-idempotent write operations unless an idempotency key exists.

Telemetry requirements

Metric / signal	Why
Function request count	Baseline usage and capacity.
SAP dependency duration	Detect downstream slowness.
SAP timeout rate	Identify retry/stability issues.
APIM failures	Detect auth, policy, or backend failures.
Foundry orchestration latency	Evaluate AI/orchestration responsiveness.
Invalid schema output rate	Catch schema contract failures.
Customer Insights journey activation failures	Detect communication-plane start issues.